

EXP – 10

nano race_counter.c

```
#include <stdio.h>
#include <pthread.h>
#include <sched.h>
```

```
#define NUM_THREADS 8
#define INCREMENTS 10000
```

```
volatile long counter = 0;
```

```
/* Thread function */
void *increment_counter(void *arg)
{
    for (int i = 0; i < INCREMENTS; i++)
    {
        /*
         * Read-modify-write operation is NOT protected.
         * Multiple threads can access counter simultaneously.
         */
        long temp = counter;

        /* Give other threads a chance to run */
        sched_yield();

        counter = temp + 1;
    }

    return NULL;
}
```

```
int main()
{
    pthread_t threads[NUM_THREADS];

    printf("=====\n");
    printf("  RACE CONDITION COUNTER PROGRAM\n");
    printf("=====\n");

    printf("Threads   : %d\n", NUM_THREADS);
    printf("Increments : %d per thread\n", INCREMENTS);
    printf("Expected   : %ld\n",
        (long)NUM_THREADS * INCREMENTS);
```

```

/* Create threads */
for (int i = 0; i < NUM_THREADS; i++)
{
    if (pthread_create(&threads[i], NULL,
        increment_counter, NULL) != 0)
    {
        perror("pthread_create");
        return 1;
    }
}

/* Wait for all threads */
for (int i = 0; i < NUM_THREADS; i++)
{
    pthread_join(threads[i], NULL);
}

printf("Actual    : %ld\n", counter);

if (counter != (long)NUM_THREADS * INCREMENTS)
    printf("Result    : RACE CONDITION DETECTED\n");
else
    printf("Result    : Expected value obtained\n");

return 0;
}

```

```

bharath@DESKTOP-6D34C2R:~/ossp_exp11$ nano race_counter.c
bharath@DESKTOP-6D34C2R:~/ossp_exp11$ gcc race_counter.c -o race_counter -pthread

bharath@DESKTOP-6D34C2R:~/ossp_exp11$
bharath@DESKTOP-6D34C2R:~/ossp_exp11$ ls -l
total 20
-rwxr-xr-x 1 bharath bharath 16312 Oct  3 09:22 race_counter
-rw-r--r-- 1 bharath bharath  1576 Oct  3 09:22 race_counter.c
bharath@DESKTOP-6D34C2R:~/ossp_exp11$ ./race_counter
=====
RACE CONDITION COUNTER PROGRAM
=====
Threads      : 8
Increments   : 10000 per thread
Expected     : 80000

Actual       : 11166
Result      : RACE CONDITION DETECTED

```

nano mutex_counter.c

```

#include <stdio.h>
#include <pthread.h>

#define NUM_THREADS 8
#define INCREMENTS 10000

long counter = 0;

/* Mutex */
pthread_mutex_t lock;

/* Thread function */
void *increment_counter(void *arg)
{
    for (int i = 0; i < INCREMENTS; i++)
    {
        /* Lock before accessing shared data */
        pthread_mutex_lock(&lock);

        counter++;

        /* Unlock after accessing shared data */
        pthread_mutex_unlock(&lock);
    }

    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];

    printf("=====\n");
    printf("    MUTEX COUNTER PROGRAM\n");
    printf("=====\n");

    printf("Threads    : %d\n", NUM_THREADS);
    printf("Increments  : %d per thread\n", INCREMENTS);
    printf("Expected    : %ld\n\n",
        (long)NUM_THREADS * INCREMENTS);

    /* Initialize mutex */
    if (pthread_mutex_init(&lock, NULL) != 0)
    {
        perror("pthread_mutex_init");
        return 1;
    }
}

```

```

/* Create threads */
for (int i = 0; i < NUM_THREADS; i++)
{
    if (pthread_create(&threads[i], NULL,
        increment_counter, NULL) != 0)
    {
        perror("pthread_create");
        pthread_mutex_destroy(&lock);
        return 1;
    }
}

/* Wait for threads */
for (int i = 0; i < NUM_THREADS; i++)
{
    pthread_join(threads[i], NULL);
}

printf("Actual    : %ld\n", counter);

if (counter == (long)NUM_THREADS * INCREMENTS)
    printf("Result    : CORRECT - MUTEX SYNCHRONIZATION\n");
else
    printf("Result    : ERROR\n");

/* Destroy mutex */
pthread_mutex_destroy(&lock);

return 0;
}

```

```

Result: CORRECT - MUTEX SYNCHRONIZATION
bharath@DESKTOP-6D34C2R:~/ossp_exp11$ nano mutex_counter.c
bharath@DESKTOP-6D34C2R:~/ossp_exp11$ gcc mutex_counter.c -o mutex_counter -pthread
bharath@DESKTOP-6D34C2R:~/ossp_exp11$ ./mutex_counter
=====
MUTEX COUNTER PROGRAM
=====
Threads      : 8
Increments   : 10000 per thread
Expected     : 80000

Actual       : 80000
Result       : CORRECT - MUTEX SYNCHRONIZATION

```

```
bharath@DESKTOP-6D34C2R:~/ossp_exp11$ echo "==== WITHOUT MUTEX ====="
./race_counter
```

```
echo ""
echo "==== WITH MUTEX ====="
./mutex_counter
==== WITHOUT MUTEX =====
=====
RACE CONDITION COUNTER PROGRAM
=====
Threads      : 8
Increments   : 10000 per thread
Expected     : 80000

Actual       : 10789
Result      : RACE CONDITION DETECTED
```

```
==== WITH MUTEX =====
=====
MUTEX COUNTER PROGRAM
=====
Threads      : 8
Increments   : 10000 per thread
Expected     : 80000

Actual       : 80000
Result      : CORRECT - MUTEX SYNCHRONIZATION
```

```
bharath@DESKTOP-6D34C2R:~/ossp_exp11$ ls -lh
total 44K
-rwxr-xr-x 1 bharath bharath 17K Oct  3 09:23 mutex_counter
-rw-r--r-- 1 bharath bharath 1.7K Oct  3 09:22 mutex_counter.c
-rwxr-xr-x 1 bharath bharath 16K Oct  3 09:22 race_counter
-rw-r--r-- 1 bharath bharath 1.6K Oct  3 09:22 race_counter.c
bharath@DESKTOP-6D34C2R:~/ossp_exp11$
```